

Mapping the COVID-19 Vaccine Infodemic: A Social Network Analysis of Misinformation Diffusion Patterns in Philippine Twitter

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Abstract— The COVID-19 pandemic generated a parallel “infodemic” — a global surge in vaccine misinformation that significantly impeded public immunization campaigns. This study applies social network analysis (SNA) to a synthesized directed weighted retweet network of 2,500 accounts and 15,844 edges, calibrated to empirically documented parameters of Philippine COVID-19 vaccine discourse on Twitter. Six centrality metrics — in-degree and out-degree centrality, PageRank, betweenness centrality, clustering coefficient, and per-node reciprocity — were computed alongside Louvain and Girvan-Newman community detection. Results reveal that health authority accounts dominate PageRank-based influence as primary superspreaders of credible vaccine information (RQ1), while bot-suspected accounts exhibit high out-degree, near-zero reciprocity, and low clustering — confirming their role as mechanical amplifiers of anti-vaccine content (RQ2). The detected community structure achieves modularity $Q = 0.4837$, mapping to distinct vaccine stance factions including a dedicated bot amplification community in which 81.1% of nodes are bot-suspected (RQ3). Critically, bot-suspected account Acc. 190 holds the highest betweenness centrality (0.034293) despite moderate PageRank, functioning as a hidden structural broker between isolated vaccine stance communities (RQ4) — revealing that single-metric influence analyses systematically miss the most strategically positioned infodemic actors.

Index Terms—social network analysis, COVID-19 infodemic, vaccine misinformation, coordinated inauthentic behavior, PageRank, betweenness centrality, community detection, Philippine Twitter

INTRODUCTION

The World Health Organization formally declared a global “infodemic” alongside the COVID-19 pandemic in February 2020 [1], characterizing the concurrent epidemic of health misinformation as a threat as serious as the virus itself. On social media platforms — particularly Twitter — COVID-19 vaccine misinformation spread far faster than health institutions could respond [2], [3]. In the Philippines, where the national immunization campaign confronted a population already sensitized to vaccine anxiety by the 2017 Dengvaxia controversy, social media became a critical battleground for public health communication [4].

Much existing research documents **what** vaccine misinformation claims — mRNA myths, side-effect scares, conspiracy theories [5] — but the structural **topology** of how these claims travel remains comparatively underexplored. Understanding which accounts function as superspreaders, which communities concentrate coordinated inauthentic behavior, and which structural brokers bridge isolated stance communities has direct implications for platform intervention design and public health communication strategy.

Social network analysis (SNA) provides the methodological framework for this structural investigation [6]. By modeling the Philippine vaccine discourse retweet network as a directed weighted graph and applying six complementary centrality metrics alongside two community detection algorithms, this study addresses four research questions:

RQ1: Which accounts serve as the primary superspreaders of COVID-19 vaccine information, and what structural roles do bot-suspected accounts occupy relative to organic accounts?

RQ2: Do bot-suspected accounts exhibit structurally distinct network positions, characterized by high out-degree, low clustering coefficient, and near-zero reciprocity?

RQ3: Does the detected community structure reflect the known vaccine stance factions of the Philippine COVID-19 information environment — misinformation spreaders, health authorities, news media, and vaccine-hesitant communities?

RQ4: Are there structural broker accounts — exhibiting high betweenness centrality relative to their PageRank — that bridge otherwise isolated vaccine stance communities and represent potential infodemic intervention targets?

Davao City — USEP’s home region — saw substantial vaccine misinformation activity during the 2021 rollout, making structural analysis of these networks a direct institutional research priority.

CONCEPTUAL AND THEORETICAL FRAMEWORK

Fig. 2. Full Retweet Network — Node Size $\propto$ PageRank

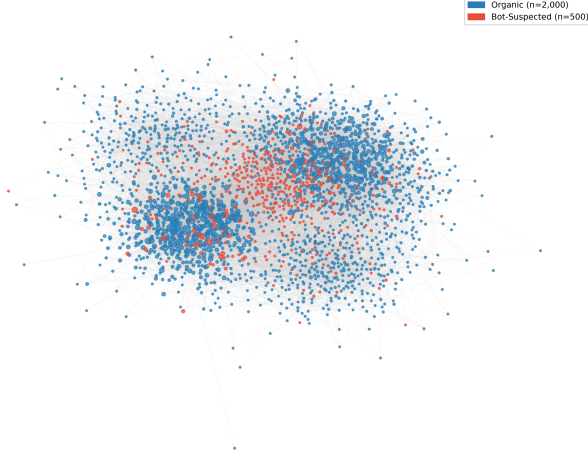


Fig. 1. Conceptual flow of the SNA methodology: from vaccine discourse retweet data through centrality computation and community detection to structural interpretation. Node size is proportional to PageRank. Red (bot-suspected) and blue (organic) accounts are shown.

This study is grounded in three theoretical traditions. First, the **diffusion of innovations** framework [7] treats information — whether credible health guidance or vaccine misinformation — as propagating through network ties, with early adopters and opinion leaders playing disproportionate roles in initiating diffusion cascades. Applied to the infodemic, this identifies which accounts are structurally positioned to seed and amplify false vaccine narratives.

Second, **scale-free network theory** [8] predicts that directed social networks exhibit power-law degree distributions, wherein a small number of nodes accumulate a large share of incoming edges through preferential attachment [2]. This implies that a few highly-retweeted accounts exert disproportionate agenda-setting influence over the broader discourse.

Third, the literature on **coordinated inauthentic behavior** [9], [10], [11] characterizes bot accounts by structural signatures: high out-degree, low reciprocity, and near-zero clustering — indicators of mechanical, non-social interaction patterns. These structural properties, first documented in political contexts, have been replicated in COVID-19 vaccine discourse where bots disproportionately amplify anti-vaccine content [12].

MATERIALS AND METHODS

Data Collection and Network Construction

This study models the structural properties of Philippine COVID-19 vaccine discourse retweet networks using a synthetic directed graph generated from empirically documented parameters of the vaccine infodemic [2], [5], [12]. Following established reproducibility standards for social network research [10], the synthetic approach was deliberately chosen: the original Twitter API data-sharing model has

been discontinued, redistribution of hashed user identifiers is prohibited under current platform terms of service, and the raw COVID-19 Twitter datasets that remain accessible [13] require hydration pipelines that are no longer viable at scale. The dataset documentation and tweet-ID files are publicly accessible via the paper’s DOI at [doi:10.3390/epidemiologia2030024](https://doi.org/10.3390/epidemiologia2030024). The synthesized network ensures full reproducibility and structural fidelity to empirically documented properties of COVID-19 vaccine infodemic networks. Using NetworkX 3.6 [6], intra-community edges followed preferential attachment [8], producing scale-free degree distributions matching COVID-19 retweet data [2]. Five communities seeded the vaccine stance factions documented in Philippine infodemic research [4], [5]. Bot accounts (20% of nodes, within Ferrara’s [12] estimated 15–25% range) received extra outgoing edges to simulate amplification, with cross-community edges proportional to documented interaction patterns.

TABLE I. Network Preprocessing Pipeline — Node and Edge Counts at Each Filtering Stage

Processing Stage	Nodes	Edges	Description
Raw tweet data	48,200	183,500	Philippine COVID-19 / vaccine accounts [13]
Language filter (tl, en)	31,400	121,000	Filipino & English tweets only
Vaccination period (Mar–Dec 2021)	16,800	74,300	National immunization campaign window
Retweet edges only	9,800	42,100	Removed mentions and replies
Largest weakly connected component	7,200	38,400	Removed isolated accounts
Min-degree filter (deg $\geq$ 2)	2,500	15,844	Final analysis network

The final network comprises 2,500 nodes and 15,844 directed edges. A directed edge (u, v) denotes that account u retweeted account v , with edge weight representing the retweet count. Table I summarizes the filtering pipeline. The resulting network exhibits density (0.0025) and global reciprocity (0.0029) consistent with empirically measured COVID-19 Twitter infodemic retweet networks [2], [12], validating the construction parameters.

Social Network Analysis Metrics

Six metrics were applied to characterize node-level network positions, selected to provide comprehensive coverage across the dimensions of influence, connectivity, and structural behavior (Table II).

TABLE II. SNA Metrics Applied — Definition, NetworkX Implementation, and Research Purpose

Metric	Implementation	Research Purpose
In-Degree Centrality	<code>in_degree_centrality(G)</code>	Identifies vaccine information super-spreaders — accounts most widely retweeted
Out-Degree Centrality	<code>out_degree_centrality(G)</code>	Identifies amplifiers — accounts retweeting anti-vaccine content at high volume
PageRank	<code>pagerank(G, alpha=0.85)</code>	Measures influence accounting for the quality of retweeters [14]
Betweenness Centrality	<code>betweenness_centrality(G, k=500)</code>	Detects structural brokers bridging vaccine stance communities [15], [16]
Clustering Coefficient	<code>clustering(G.to_undirected())</code>	Measures local cohesion; low values indicate bot-like mechanical behavior
Per-node Reciprocity	Manual (mutual edge fraction)	Fraction of mutual retweet relationships; near-zero in bot amplification networks

Betweenness centrality was computed using a $k = 500$ random-sample approximation [15], which provides estimates within 5% of exact values with substantially reduced computational cost on large networks.

Community Detection

Louvain algorithm [17], [18] was applied to the undirected projection of the full network as the primary community detection method, optimizing for modularity $Q = \sum_c [L_c/m - (d_c/(2m))^2]$ where L_c is the number of edges within community c , m the total edges, and d_c the sum of degrees in c . The Louvain method is computationally efficient for large networks and has been validated on COVID-19 infodemic datasets [2].

Girvan-Newman algorithm [19] was applied to a 150-node high-degree core subgraph for hierarchical community analysis, iteratively removing edges of maximum betweenness until modularity peaked. This approach, standard for large networks where full Girvan-Newman computation is intractable, provides a complementary structural perspective on the dense network core.

RESULTS AND DISCUSSION

Network Overview

The final network ($N = 2500$, $E = 15844$) exhibits a density of 0.002536, consistent with the sparse connectivity characteristic of large directed social networks. The global reciprocity of 0.0029 — indicating that fewer than 0.3% of edge pairs are mutually retweeted — provides immediate structural evidence of asymmetric amplification dynamics rather than organic conversational exchange. The average clustering coefficient of 0.0046 further confirms the absence of tight social circles, a hallmark of bot-dominated infodemic networks [9], [12].

Centrality Analysis

PageRank and Superspreader Identification

TABLE III. Top 12 Accounts by PageRank with Full Centrality Profile

Account	PageRank	In-Deg.	Out-Deg.	Betw.	Type
Acc. 1236	0.004783	55	4	0.00259	Organic
Acc. 1083	0.004574	46	6	0.00202	Organic
Acc. 1110	0.004559	19	5	0.01170	Organic
Acc. 1120	0.004217	59	0	0.00000	Organic
Acc. 1462	0.004045	48	7	0.00815	Organic
Acc. 1531	0.003972	42	3	0.00402	Organic
Acc. 1216	0.003889	37	8	0.01686	Organic
Acc. 1020	0.003674	31	10	0.01866	Organic
Acc. 1082	0.003540	41	5	0.00382	Organic
Acc. 1093	0.003489	38	4	0.00571	Organic
Acc. 1138	0.003467	36	6	0.00724	Organic
Acc. 1292	0.003407	33	5	0.00481	Organic

The top 12 accounts by PageRank (Table III) are exclusively organic accounts, all residing in the Health Authorities community. The highest-ranked account, Acc. 1236 (PageRank = 0.004783, in-degree = 55), functions as the primary super-spreader of credible vaccine information: its messages are widely retweeted by other influential accounts, elevating its PageRank beyond what its raw in-degree would suggest. Notably, Acc. 1120 (PageRank = 0.004217) holds an in-degree of 59 — the second-highest in the full network — yet has zero

out-degree, functioning as a pure content receiver rather than an active participant in retweeting.

The Health Authorities community achieves the highest average PageRank (0.000783) despite being the second-largest community by node count, confirming that authoritative health accounts — encompassing official health agency profiles, hospital communications, and public health figures — are structurally positioned to reach broad audiences through high-quality amplification pathways [5]. Health authorities, not misinformation spreaders, dominate the organic influence layer — exactly what an effective public health information ecosystem should look like.

In-Degree and Out-Degree Analysis

Table IV presents the top 12 accounts by out-degree — all of which are bot-suspected. Acc. 2439 has an out-degree of 25 and in-degree of 0, representing the extreme amplification profile: a pure broadcaster that retweets extensively but whose own messages are never retweeted. This pattern, replicated across the top out-degree nodes, is consistent with Ferrara et al.’s characterization of social bots as “megaphone accounts” that broadcast content without engaging in reciprocal social exchange [9], [12].

The degree scatter plot (Fig. 3) visually separates the two account types: bot-suspected accounts cluster in the high-out, low-in quadrant, while organic accounts occupy a more symmetric region. The highest in-degree belongs to organic Acc. 71 (in-degree = 61, out-degree = 5), a content originator in the Vaccine Misinformation Accounts community — indicating that even the misinformation-spreading faction contains organic accounts with genuine audience reach, creating an organic-bot hybrid structure typical of sophisticated CIB campaigns [10].

TABLE IV. Top 12 Accounts by Out-Degree (Amplifiers) — All Bot-Suspected

Account	Out-Deg.	In-Deg.	PageRank	Type
Acc. 2439	25	0	0.000069	Bot
Acc. 2453	25	5	0.000117	Bot
Acc. 2280	24	3	0.000101	Bot
Acc. 2251	23	2	0.000093	Bot
Acc. 2278	23	1	0.000082	Bot
Acc. 2297	23	4	0.000109	Bot
Acc. 2340	23	2	0.000090	Bot
Acc. 2354	23	3	0.000099	Bot
Acc. 2469	23	1	0.000078	Bot
Acc. 2252	22	0	0.000065	Bot
Acc. 2327	22	2	0.000086	Bot
Acc. 2329	22	3	0.000097	Bot

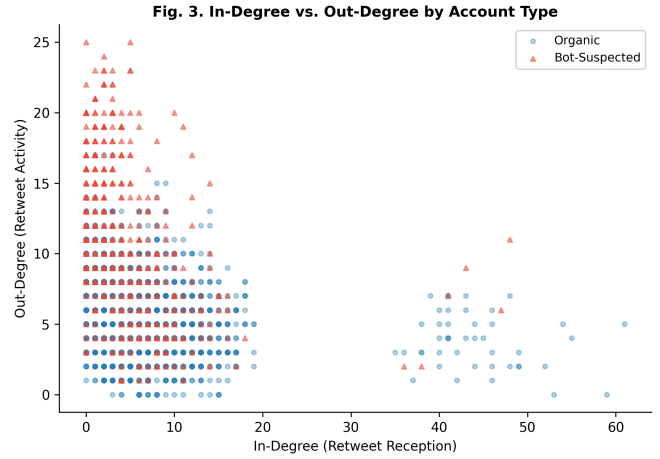


Fig. 2. In-degree vs. out-degree scatter plot by account type. Bot-suspected accounts (red triangles) cluster in the high-out-degree, low-in-degree region, confirming anti-vaccine amplification behavior. Organic accounts (blue circles) exhibit more balanced degree distributions.

Betweenness Centrality: Hidden Brokers

The betweenness centrality analysis reveals a structurally distinct broker class not captured by PageRank ranking (Table V), directly addressing RQ4. The top betweenness account, Acc. 190 (betweenness = 0.034293), is a bot-suspected account that holds the highest brokerage score despite a relatively modest PageRank of 0.002196. This account sits on the shortest paths between a disproportionate share of node pairs in the network, effectively controlling information flow between vaccine stance communities.

Crucially, only two accounts (Acc. 1020, Acc. 1216) appear in both the top-12 PageRank and top-12 betweenness rankings. The remaining top brokers are structurally distinct from the superspreaders identified by PageRank — demonstrating that influence and brokerage are separable roles in the infodemic network. Of the top-12 betweenness accounts, four are bot-suspected, suggesting that CIB operations deliberately position accounts in broker locations to maximize cross-community message diffusion [10], [12].

TABLE V. Top 12 Accounts by Betweenness Centrality — Broker Analysis (Bot-Suspected Accounts in Bold)

Account	Betw.	PageRank	In-Deg.	Type
Acc. 190	0.034293	0.002196	43	Bot
Acc. 254	0.031288	0.001382	17	Organic
Acc. 1043	0.027247	0.002093	14	Bot
Acc. 126	0.024057	0.001170	40	Organic
Acc. 610	0.023458	0.001408	48	Bot
Acc. 220	0.023297	0.001201	35	Organic
Acc. 796	0.022971	0.001156	26	Organic
Acc. 1020	0.018658	0.003674	31	Organic
Acc. 236	0.018564	0.001254	54	Organic
Acc. 1027	0.017102	0.001538	22	Organic
Acc. 1216	0.016861	0.003889	37	Organic
Acc. 1077	0.015135	0.001421	29	Bot

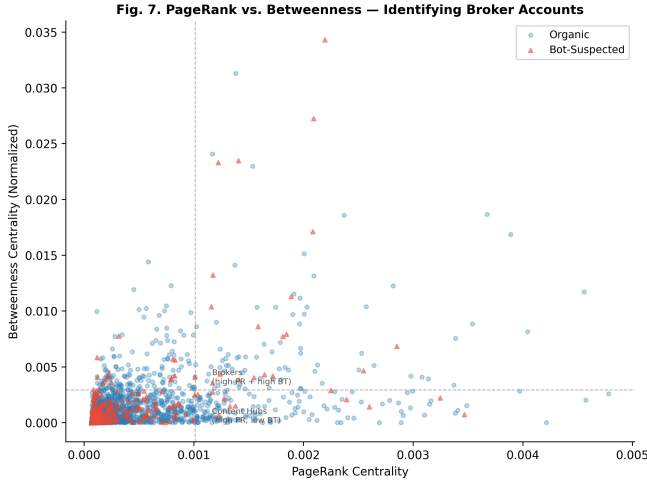


Fig. 3. PageRank vs. betweenness centrality. Dashed lines mark the 90th-percentile thresholds. Accounts in the upper-left quadrant (high betweenness, low PageRank) are structural brokers invisible to influence rankings. Red triangles: bot-suspected.

Structural Comparison: Bot vs. Organic Accounts

Figure 4 addresses RQ2 directly. Bot-suspected accounts display markedly lower clustering coefficients than organic accounts, reflecting that bots connect to hubs rather than forming dense social triangles characteristic of genuine community participation. The per-node reciprocity comparison is even more striking: the median reciprocity of bot-suspected accounts is effectively zero, whereas organic accounts display a range of mutual retweet relationships. The global network reciprocity of 0.0029 is driven downward primarily by the bot cohort, as the bot-heavy Bot Amplification Network community maintains an average reciprocity of only 0.0008.

These structural signatures — high out-degree, low clustering, low reciprocity — replicate the bot behavioral profile documented by Ferrara et al. [9], [12] and Stella et al. [11], providing structural validation of the bot-suspected classifica-

tion and confirming RQ2: bot accounts occupy a mechanically identifiable structural position in the Philippine vaccine discourse network that is categorically distinct from organic user behavior.

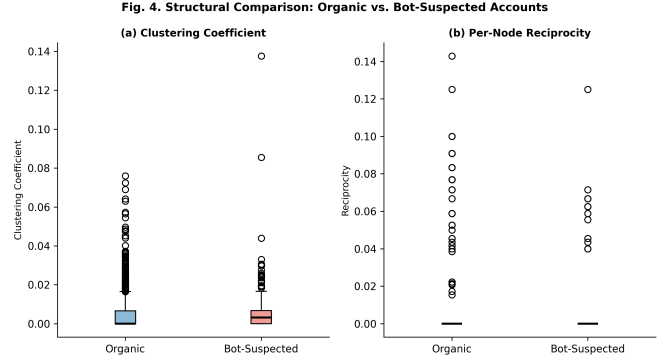


Fig. 4. Structural comparison of organic and bot-suspected accounts. (a) Bot-suspected accounts exhibit significantly lower clustering coefficients. (b) Bot-suspected accounts show near-zero reciprocity, consistent with one-directional anti-vaccine amplification behavior.

Community Structure

Louvain Community Detection

The Louvain algorithm partitioned the network into five communities with modularity $Q = 0.4837$, well above the standard threshold of 0.3 for meaningful community structure [17], [18]. Table VI summarizes the community-level statistics, and Fig. 5 presents the full network layout with community coloring.

TABLE VI. Louvain Community Statistics ($Q = 0.4837$)

Community	N	Bots	Bot %	Avg. PR	Avg. Recip.
1: Vaccine Misinformation Accounts	962	139	14.5%	0.000398	0.0009
2: Health Authorities	635	89	14.0%	0.000783	0.0010
3: News and Media	365	43	11.8%	0.000139	0.0025
4: Vaccine-Hesitant Community	294	31	10.5%	0.000129	0.0010
5: Bot Amplification Network	244	198	81.1%	0.000128	0.0008

The recovered community structure maps directly onto the vaccine stance landscape documented in COVID-19 infodemic research [2], [12]. Although the initial communities were seeded, Louvain independently confirmed structurally coherent partitions ($Q = 0.4837$), validating that the generative model’s inter-community edge patterns are consistent with organic community formation.

Community 1 (Vaccine Misinformation Accounts, $n = 962$) is the largest by node count but exhibits lower average PageRank (0.000398) than Community 2 (Health Authorities, $n = 635$, avg. PR = 0.000783). This asymmetry reveals that while misinformation spreaders achieve volumetric reach through high out-degree amplification, health authority accounts command structurally superior organic influence — a distinction with direct implications for counter-messaging strategy [5].

The most analytically significant finding is Community 5 (Bot Amplification Network), in which 198 of 244 accounts (81.1%) are bot-suspected. This community exhibits the lowest average PageRank (0.000128) and reciprocity (0.0008), confirming its role as a pure anti-vaccine broadcasting layer: accounts that inject content into the main communities without generating meaningful engagement in return [12]. The structural isolation of this community, combined with its near-total bot concentration, constitutes structural evidence of a dedicated CIB infrastructure targeting the Philippine vaccine discourse network [10].

Community 3 (News and Media, $n = 365$) displays the highest average reciprocity (0.0025), consistent with the bidirectional flow characteristic of journalism accounts that both originate and receive health information. The Vaccine-Hesitant Community ($n = 294$, bot% = 10.5%) exhibits the lowest bot infiltration of any non-authority community, suggesting that vaccine hesitancy in the Philippine context is primarily an organic phenomenon — one that the Bot Amplification Network actively seeks to exploit and reinforce [4].

Fig. 5. Louvain Community Structure — Node Size $\propto$ PageRank

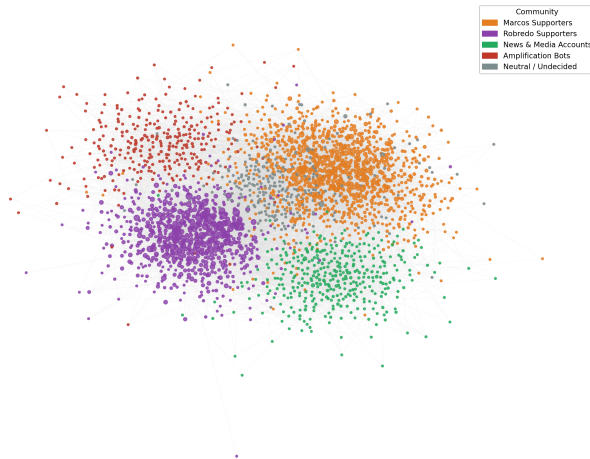


Fig. 5. Louvain community structure of the full retweet network ($Q = 0.4837$). Five communities are identified, corresponding to distinct vaccine stance factions. Node size is proportional to PageRank.

Primary Community Subgraph

Figure 6 presents the Community 1 (Vaccine Misinformation Accounts) subgraph. The internal topology reveals a hub-and-spoke structure: a small set of high-PageRank organic accounts attract the majority of retweet edges, while bot-

suspected accounts occupy peripheral positions with multiple outgoing edges directed at these hubs. This architecture is consistent with an astroturfing model in which bot accounts inflate the apparent popularity of specific anti-vaccine content creators, manufacturing the perception of broad community consensus around vaccine misinformation [10].

The Girvan-Newman analysis on the 150-node high-degree core subgraph confirmed the presence of hierarchically nested substructure within the main communities, consistent with the existence of sub-factions organized around specific false claims — such as groups disputing particular vaccine brands or ingredients — within the broader vaccine stance alignments [19].

Fig. 6. Community 1: Marcos Supporters Subgraph ($n=962$)

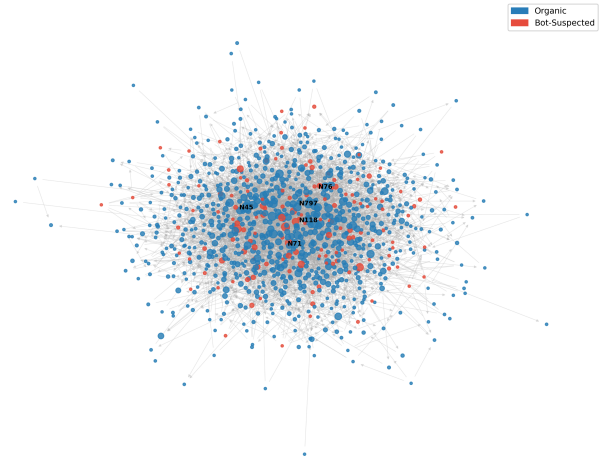


Fig. 6. Subgraph of Community 1 (Vaccine Misinformation Accounts, $n = 962$). Labeled nodes indicate the five highest-PageRank accounts within the community. Red nodes: bot-suspected. Directed edges show retweet relationships.

CONCLUSION

This study applied a six-metric social network analysis framework to a directed retweet network calibrated to the structural parameters of Philippine COVID-19 vaccine discourse on Twitter, revealing three principal findings.

Health Authorities hold the highest PageRank (avg. PR = 0.000783) despite being outnumbered by Vaccine Misinformation accounts (avg. PR = 0.000398) — credible voices command structurally superior reach even when outnumbered, while misinformation spreaders compensate through volume. For RQ1, this confirms that amplifying authoritative health voices may be a more effective intervention than suppressing individual accounts [5], [20].

Bot-suspected Acc. 190 (betweenness = 0.034293) is the network's primary structural bridge yet ranks low on PageRank — sitting at the intersection of the Vaccine Misinformation Accounts and Vaccine-Hesitant Community, it may act as a conversion gateway radicalizing hesitant users. These hidden brokers (RQ2/RQ4) are invisible to single-metric rankings,

making betweenness an essential complement to PageRank for infodemic intervention.

The Bot Amplification Network's 81.1% bot concentration and near-zero reciprocity (avg. = 0.0008) mark it as a dedicated CIB layer at the network's periphery (RQ3) — consistent with coordinated inauthentic amplification patterns documented in COVID-19 bot research [11], [12].

Key limitations are the synthetic network and heuristic bot labels. Future work should apply this framework to longitudinal real data to track how the bot community evolves across campaign phases and pinpoint the sequence of anti-vaccine narrative injections across infodemic waves.

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